Industrial Visual Anomaly Detection: A Technical Case Study

Executive summary · independent project on public benchmark images (VisA), not factory data

Technical feasibility on public data only. Programmatic metric verification completed. Human owner review pending.

Problem and prototype

On a production line, defects are rare and varied, so labelled defect examples are scarce. The question: can a model that has only seen good boards send the right circuit-board images to a human reviewer, without burying the reviewer in false alarms?

The prototype uses frozen pretrained CNN features and a per-position Gaussian (Mahalanobis) model of normal appearance, registers each board, and flags an image for review when its score exceeds a threshold frozen beforehand on separate normal images. With training on normal images only, VisA defect labels and masks are reserved for evaluation. It recommends a review and never rejects a board.

GO / NO-GO

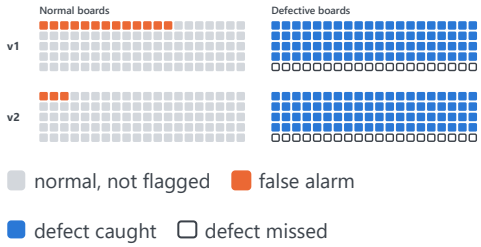
Not ready for production-affecting deployment. Although v2 reduced false alarms, it did not meet the proposed sensitivity and robustness requirements. Rule-based verdict against unchanged criteria: INCONCLUSIVE.

Proposed next step: bounded, non-disruptive shadow-mode validation, subject to site requirements, authorization and entry checks. Factory access has not been obtained.

Controlled improvement on PCB2, a board type not used in development

v1 → v2: one pre-registered change — score only inside the board region. 100 defective + 100 normal test images, thresholds frozen before testing, evaluated once; 95% confidence intervals.

	v1	v2	Change, v2 − v1
Defect recall	80%	**80%**	unchanged
False alarms, test normals	13%	**3%**	−10 pts (CI −17 to −3)
False alarms, holdout normals	5.3%	**4.0%**	−1.3 pts (CI −4.6 to +2.0)
AUROC	0.927	**0.979**	+0.052 (CI +0.028 to +0.082)



Fewer false alarms and better ranking; recall did not improve and stays below the proposed 90% target. Median latency per image (GPU): v1 194 ms, v2 224 ms.

Follow-up finding on PCB3, a separate board type

A pre-registered global brightness/contrast correction met its own decision rule ("ROBUSTNESS IMPROVEMENT SUPPORTED"; darkening difference −4.7 pts, CI −8.6 to −0.8), but that result is **weakly informative**, and the correction made a mild blur raise false alarms by +14.5 pts. **Decision: do not adopt it.** The uncorrected pipeline is not production-ready either: brightening raised its false alarms by +15.2 pts. PCB3 numbers are not comparable with PCB2's.

Disclosure: after PCB3 test scoring, one input check in the analysis program was repaired (7dbd302); no score, threshold or decision changed. Full record in the repository.

Remaining risks

- **Detection:** recall 80% against a proposed 90%; the smallest third of defects only 56%.
- **Lighting:** a 10% darker image raised PCB2 false alarms from 3.6% to 28.7%.
- **Portability:** on a green background the original board-finding rule marked 96% of the frame as board, silently.
- **Evidence base:** benchmark images and synthetic perturbations; targets are the developer's proposal. No end user, factory image, pilot or financial return.

Next recommended validation step

- Confirm site requirements with the quality owner: cost of a missed defect versus one review, review capacity, defects that matter.
- Run the unchanged tool in shadow mode on real line images, affecting no decision.
- Entry gates: board segmentation and registration on the site's own normal images; a per-image lighting check; a labelled sample of real defects, with recall reported by defect size.
- Decide Go / No-Go again on that evidence.

Data: VisA dataset, Zou et al., ECCV 2022, CC BY 4.0 (github.com/amazon-science/spot-diff). Pretrained network: torchvision Wide-ResNet-50-2 (ImageNet weights). PCB1–PCB3 are benchmark board types, not manufacturing sites. Not affiliated with or endorsed by the dataset authors or any company.

Built with AI-assisted development; project-owner technical review pending. Project owner: Yu-Chen (Leon) Lai. Protocols and raw results: private development repository. Extended brief: website download.

Website link pending first deployment.